

RESEARCH MONOGRAPH

XJTU
SCHOOL OF MATHEMATICS AND STATISTICS

Deep Learning & Computer Vision Notes

A Comprehensive Guide to Deep Learning and
Computer Vision

TW

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Overview

1.1 Sample

Definition 1.1.1 ▸ TW Notes

twnotes is a professional L^AT_EX template based on amzn_{otes}, customized for research notes.

Theorem 1.1.2 ▸ Cool Theorem

This L^AT_EX template is cool.

Proof. Here is what some math looks like:

$$\forall (n \in \mathbb{N}) \left[\sum_{i=1}^n i = \frac{n(n+1)}{2} \right]$$

$$\mathcal{L}\{f\}(s) = \int_0^{\infty} e^{-st} f(t) dt$$

Some more math:

$$\lfloor x \rfloor = \max\{n \in \mathbb{Z} : n \leq x\}$$

$$\lceil x \rceil = \min\{n \in \mathbb{Z} : n \geq x\}$$

The math is good. ■

As seen in [Cool Theorem](#) (Theorem 1.1.2), this template is pretty cool.

1.2 Installation

To install this package, simply add the `twnotes.cls` file to the same directory as the `.tex` file you want to use it with.

1.3 Usage

Once installed, you can use the `twnotes` document class with `\documentclass{twnotes}`.

Code Snippet 1.3.1 ▸ Using twnotes

```
1 \documentclass[math,code]{twnotes}
```

The `math` option enables `thmbox` and various math macros. The `code` option enables `codebox` and code environments.

Using the boxes is pretty straight-forward. They're based on the `tcbtheorem` from `tcolorbox`. We can see how to use one below:

Code Snippet 1.3.2 ▸ Using boxes

```
1 \begin{exbox}{Cool Example}{coolexample}
2   This is my example!
3 \end{exbox}
4 \nameref{ex:coolexample} (example \ref{ex:coolexample}) is pretty
   ↪ cool.
```

This is what Code Snippet 1.3.2 produces:

Example 1.3.3 ▸ Cool Example

This is my example!

Cool Example (example 1.3.3) is pretty cool.

`twnotes` provides many predefined color boxes. The ones that are numbered include:

- `dfnbox` – definition box
- `exbox` – example box
- `tecbox` – technique box
- `thmbox` – theorem box (with option `math`)
- `codebox` – code box (with option `code`)

Some unnumbered/unindexed boxes include:

- `genbox` – general info box

- `notebox`

You can also easily create your own boxes with the `twnotes` style using the `\newtwbox` command in the preamble. These will be numbered and indexed like the other boxes.

Code Snippet 1.3.4 ▸ Using `\newtwbox`

```
1 \newtwbox{environment name}{title}{ref label}{color}[tcolorbox style]
```

Really really long chapter title that has to use more than one line because it is so long

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